# Ian Dahlin

Lead Software Engineer

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## SUMMARY

Lead Software Engineer directing the architecture and development of desktop applications, automation platforms, and simulation tools. Experienced leading teams to deliver scalable, maintainable software. B.S. in Computer Science (AI/ML); published at IEEE WACV 2026.

## EXPERIENCE

**Brooks Building Solutions** — Jacksonville, FL

Lead Software Engineer — Nov 2025 – Present

- Lead an engineering team building a C#/.NET desktop app for engineering deliverables.
- Architected selection-driven generation of HVAC unit schematic and wiring diagrams, bills of materials, parts and points lists, and sequences of operation.
- Architect engineering-grade simulation tools for building system design and validation.

Software Engineer — Jul 2025 – Nov 2025

- Built internal tools and web applications that automate company processes.
- Built reusable diagram tools and templates that reduced manual HVAC design work.

**University of North Florida** — Jacksonville, FL

Machine Learning Research Assistant — Aug 2024 – May 2025

- Developed and trained 3D object classification models (PointNet, PointNet++, and variants) on LiDAR point clouds, achieving 91% classification accuracy.
- Co-developed TopoRec, a topological data analysis method for point-cloud recognition.
- Built preprocessing pipelines with downsampling that cut computational load by 50%.

## PUBLICATIONS

**TopoRec: Point Cloud Recognition Using Topological Data Analysis**

*IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2026 | Co-author

## EDUCATION

**University of North Florida** — Jacksonville, FL

Bachelor of Science in Computer Science — May 2025

- Concentration: Artificial Intelligence & Machine Learning

## PROJECTS

**AeroAtlas – AI Travel Planner** — *Audience Choice Award, UNF Computing Symposium*

- Built a fully working web app with a five-person team that uses LLMs to generate custom trip plans with activities, images, and local insights.
- Integrated real-time data from multiple third-party APIs.

## SKILLS

**Languages:** C#, Java, Python, C, SQL, MATLAB, XAML, HTML, CSS
**Frameworks & Libraries:** .NET, React, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Open3D
**Tools & Platforms:** Git, GitHub, AutoCAD, Blender, Linux, Unix, LiDAR Processing